

Initial report: Discrete Diffusion for Anatomical Shape Generation

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Project repository: github.com/Marodriguezfu/PRIM-Discrete-Difusion-for-Anatomical-Shape-Generation

1 Project overview and objectives

This PRIM project, "*Discrete Diffusion for Anatomical Shape Generation*", studies if vector-quantised autoencoder models can learn a useful discrete 3D representation of the prostate anatomy from MRI scans. In this first phase, we adapt a 3D VQ-UNet architecture with a vector-quantisation bottleneck [1] to segment volumetric T2-weighted prostate MRI data from the PROSTATE-DIAGNOSIS and Prostate-3T collections [2, 3]. The main objective is to obtain stable and anatomically consistent three-class segmentations (background, transition zone, peripheral zone) and to learn a reliable codebook of discrete tokens. In the next phase, these tokens will be used as the input space for a discrete diffusion model [4] that aims to generate realistic anatomical shapes.

2 Baseline model and dataset

The starting point of this project is the public repository *Vector-Quantisation-for-Robust-Segmentation* [5]. This code implements a 3D VQ-UNet model for prostate MRI segmentation. In the original version, the task is **binary**: all prostate voxels (transition zone and peripheral zone) are merged into one foreground class, and all other voxels are background. The code reports three main metrics for binary segmentation: Dice coefficient, Hausdorff distance (HD) and average surface distance (ASD).

The repository also provides a complete data loading pipeline for 3D medical images. It performs intensity normalisation, resampling to a common voxel size, and cropping or padding to a fixed volume size. Simple geometric data augmentations are applied during training. In the baseline configuration, all labels are converted to a single binary prostate mask before training.

The MRI volumes used in this project come from the PROSTATE-DIAGNOSIS and Prostate-3T collections on TCIA [2, 3]. These datasets provide T2-weighted axial prostate images with expert manual contours for the central gland and the peripheral zone. In the original repository, the training and validation cases are taken only from the BMC centre, while the test set uses only cases from the RUNMC centre. This centre-based split can introduce a domain bias, because the acquisition protocols may be different in the two hospitals and the anatomical characteristics of the cases are not explicitly balanced.

To obtain a more fair and stable evaluation protocol, the dataset was reorganised using a custom script that creates a new split. The script reads the available CSV files, computes the transition zone (label 1) and peripheral zone (label 2) volumes for each case, and assigns a patient identifier from the file path. It then builds volume-based bins for TZ and PZ and performs a stratified split by patient, with an approximate 55/12/12 ratio for train, validation and test. The final `train.csv`, `validation.csv` and `test.csv` files therefore contain a mix of BMC and RUNMC cases, balanced by TZ/PZ volumes and grouped by patient. This balanced split is used for all experiments in this report.

3 Current method and pipeline modifications

The current model is called VQUNet3Dposv3. It is a 3D U-Net with a vector quantisation bottleneck, inspired by the Neural Discrete Representation Learning work [1]. The network takes a full prostate MRI volume as input and predicts three classes: background, transition zone (TZ) and peripheral zone (PZ). In the encoder, 3D convolutions and residual blocks extract features at different scales. At the deepest level, these features are projected to an embedding space of size 256 and passed through a vector quantiser with a codebook of 1024

entries. Each latent vector is replaced by its nearest codebook entry, which creates a grid of discrete tokens in 3D. The decoder then upsamples these tokens and combines them with skip connections to reconstruct the multiclass segmentation mask. Compared to the original binary repository [5], the internal Transformer block was removed to reduce memory usage, and a lighter 3D positional encoding module was added to give the model basic spatial information without a large cost in GPU memory.

Several changes were also made to the data pipeline. Preprocessing is now standardised for training, validation and testing. All volumes are resampled to a common voxel spacing, cropped or padded to a fixed size of $(192, 192, 64)$, and normalised in intensity. Simple 3D data augmentation is applied during training to increase robustness. The masks are kept in three-class format (background, TZ, PZ) and converted to one-hot encoding only inside the training loop, instead of collapsing TZ and PZ into a single foreground label as in the original code [5]. The dataset split is the balanced 55/12/12 train/validation/test partition described in the previous section, and it is used consistently in all experiments.

The training and logging code was also updated. The calculation of the main evaluation metrics (Dice, Hausdorff distance and average surface distance) now uses well-tested utilities from MONAI. The metrics are computed per volume, without the background class, and then averaged over the full validation set. The validation loop was corrected to reset the metric accumulators properly and to track both the best Dice and a weighted multi-metric score. Checkpoints are saved with filenames that include the epoch and key metric values, and the training curves are exported in EPS format for easy inclusion in scientific reports.

4 Experimental setup

All experiments use the VQUNet3Dposv3 model for 3D prostate MRI segmentation, trained on the PROSTATE-DIAGNOSIS and Prostate-3T datasets [2, 3]. The network follows the vector-quantised autoencoder idea from van den Oord et al. [1] and is adapted from the public repository of Ainkaran Santhi [5]. Each volume is resampled and cropped to a fixed size of $192 \times 192 \times 64$ voxels with a voxel spacing of $0.5 \times 0.5 \times 1.5$ mm. We train with a batch size of 1, using the AdamW optimizer with learning rate 1×10^{-4} and weight decay 1×10^{-5} . The loss is a sum of Dice and Cross-Entropy, which helps to deal with the imbalance between background, transition zone, and peripheral zone. For the three-class model we test two training lengths, 50 and 100 epochs, and we change how often we run validation: either every epoch or every 5 epochs. For model selection we first use only the Dice score, and then a simple multi-metric score that combines Dice, HD, and ASD with weights (w_D, w_{HD}, w_{ASD}) chosen to give more importance to Dice while still encouraging good boundary accuracy [4].

Table 1: Main experimental settings for the three-class prostate segmentation model.

Item	Value
Input volume size	$192 \times 192 \times 64$ voxels
Voxel spacing	$0.5 \times 0.5 \times 1.5$ mm
Batch size	1 volume
Optimizer	AdamW, lr = 1×10^{-4} , weight decay = 1×10^{-5}
Loss	Dice + Cross-Entropy
Epochs (3-class runs)	50 and 100
Validation frequency	every epoch (check_val = 1) or every 5 epochs (check_val = 5)
Multi-metric weights (50 epochs)	$w_D = 1.0$, $w_{HD} = 0.0$, $w_{ASD} = 0.0$
Multi-metric weights (100 epochs)	$w_D = 8.0$, $w_{HD} = 1.0$, $w_{ASD} = 1.0$

5 Key quantitative results

Table 2 shows the main quantitative results for prostate segmentation. We compare the original binary baseline model from the public repository of Ainkaran Santhi [5] and the proposed three-class VQUNet3Dposv3 model, based on vector-quantised autoencoders [1]. For each setting, we report the mean and standard deviation of the Dice index, Hausdorff distance (HD) and average surface distance (ASD) on the balanced test split.

The binary baseline model reaches a good overlap between prediction and ground truth, with a mean Dice of about 0.75. However, its boundary errors are very large: the mean HD is close to 37.7 mm and the mean ASD is about 6.7 mm. This happens even though the task is easier, because the model only needs to separate

Table 2: Mean $\pm$ standard deviation of Dice, Hausdorff distance (HD) and average surface distance (ASD) for the binary baseline and the proposed three-class model on the balanced test split.

Setting	Dice	HD [mm]	ASD [mm]
Baseline (binary, 50 epochs)	0.7542 ± 0.0664	37.6952 ± 15.0929	6.7354 ± 3.3346
3-class, 50 epochs (best Dice)	0.7407 ± 0.0772	18.4319 ± 6.1275	2.0209 ± 0.6272
3-class, 100 epochs (best multi-metric)	0.7729 ± 0.0508	16.7430 ± 6.5931	1.8607 ± 0.4019

the prostate from the background and does not distinguish between transition zone and peripheral zone.

The three-class model trained for 50 epochs already gives a similar Dice score (about 0.74), but it strongly reduces the boundary errors. The mean HD falls to around 18.4 mm and the mean ASD to about 2.0 mm. This means that the three-class VQ-based model produces smoother and more anatomically plausible contours, even if the segmentation problem is more difficult.

When we train the three-class model for 100 epochs and select the checkpoint with the best multi-metric score, all metrics improve again. The mean Dice increases to about 0.77, and both HD and ASD are further reduced (around 16.7 mm and 1.86 mm, respectively). This shows that a longer training and a combined objective on Dice, HD and ASD help the model to find a better balance between region overlap and precise boundaries.

Additional 100-epoch experiments with a different validation schedule give very similar results, with only a small drop in performance. This suggests that the training process of the three-class VQUNet3Dposv3 model is stable and robust across different choices of validation frequency.

6 Technical challenges and next steps

During this first phase, we faced several technical challenges, mainly related to GPU memory. The original repository by Ainkaran Santhi [5] includes an internal Transformer block and a larger codebook, which are hard to train on 3D prostate volumes with the available hardware. To reduce memory use, we removed the Transformer module, fixed the batch size to 1 volume, and chose a smaller vector-quantisation codebook, following the general ideas of Neural Discrete Representation Learning [1]. These changes allowed the model to train in a stable way, without running out of memory, while still keeping good segmentation performance.

We also had to reorganise the dataset and correct the way metrics and validation were computed. The initial centre-based split and some parts of the validation loop could give a biased or unstable estimate of performance on the PROSTATE-DIAGNOSIS and Prostate-3T data [2, 3]. To solve this, we created a new balanced split by patient and by TZ/PZ volume, and we rewrote the validation code so that Dice, HD and ASD are computed consistently and averaged over the full validation set. This new protocol is now used for all experiments and gives more reliable comparisons between different runs.

Several open questions remain for the next steps. First, the stability of the codebook is not fully explored: we still need to study how different sizes of the dictionary and different values of the commitment loss affect the quality of the learned tokens. Second, once the segmentation model is fully validated, we may try to reintroduce some form of attention, in a lighter way, to see if it brings extra gains without breaking the memory budget. Finally, the main goal of the project is to build a discrete diffusion model on top of the learned tokens [4]. In the next phase, we plan to use the 3D codebook from VQUNet3Dposv3 as the latent space where a discrete diffusion process will learn to generate new, realistic anatomical shapes of the prostate.

References

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